Cyrus Yong

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Education

Georgia Institute of Technology — Atlanta, GA

B.S. Computer Science & B.S. Electrical Engineering | *Minor: Robotics* | *GPA: 3.5/4.0* — *Aug 2024 – May 2028*

- EE Concentrations: Robotics & Autonomous Systems, Circuit Technology | Dean's List: Fall 2024, Spring 2025

Georgia State University — Atlanta, GA

B.S. Computer Science (Transferred to Georgia Tech) | *GPA: 3.7/4.0* — *Aug 2022 – May 2024*

- President's List: Spring 2023, Spring 2024 | Dean's List: Fall 2023

Experience

Intelligent Platform for Crowdsourcing (VIP) — GitHub — Atlanta, GA

Undergraduate Researcher, Georgia Tech — *Aug 2025 – May 2026*

- Built a structured data extraction pipeline using LiteLLM on GT's PACE GPU cluster, parsing unstructured input into defined fields for downstream workflows.
- Developed a RAG chatbot backend with controlled retrieval from curated resources, reducing hallucinations and improving system reliability.
- Designed a React + FastAPI full-stack application with custom document intake (BeautifulSoup, PyPDF) for user customized knowledge base.

Projects

STM32 Vending Machine — Embedded Control System — GitHub — June 2026 – Present

C, STM32 Nucleo-F446RE (ARM Cortex-M4), FSM, PWM Servos, GPIO EXTI, I2C, SPI/FatFS

- Building C firmware on an STM32 Nucleo-F446RE (ARM Cortex-M4) driven by a finite state machine that handles push button navigation (GPIO EXTI), payment logic, and PWM control of MG995 servo gates across three drink stacks.
- Extending the design with IR break beam sensors for coin detection and drop confirmation, an SSD1306 OLED menu (I2C), and DS3231 RTC timestamped transaction logging to microSD over SPI (FatFS).

STM32 Temperature Data Logger — Embedded Firmware — GitHub — June 2026 – Present

C, STM32 Nucleo-F446RE (ARM Cortex-M4), BME280 (I2C), SPI/FatFS

- Developing C firmware on an STM32 Nucleo-F446RE (ARM Cortex-M4) to sample a BME280 sensor (temperature, humidity, pressure) over I2C at customizable intervals.
- Logging timestamped readings to a FAT32 microSD card over SPI (FatFS) for offline retrieval and analysis, implementing peripheral drivers and buffering on bare metal.

USB HID Macropad — Custom Firmware & PCB — GitHub — May 2026 – Present

C (bare-metal), Raspberry Pi Pico (RP2040), TinyUSB, KiCad, 3D Printing

- Developing bare metal C firmware on the RP2040 that enumerates as a USB HID composite device, handling descriptor setup and key event reporting through the TinyUSB stack.
- Programmed each key to emit a predefined multi keystroke macro over USB HID, with a custom KiCad PCB and custom 3D printed enclosure, building on a prior single button HID controller.

USB HID Autoclicker — Custom Firmware & PCB — May 2026 - June 2026

C (bare-metal), Raspberry Pi Pico (RP2040), TinyUSB, KiCad

- Built a bare metal C firmware project from scratch on the RP2040, emulating a USB HID mouse over the TinyUSB stack and designing a first revision custom PCB in KiCad.

LTC2308 ADC Peripheral for SCOMP — ECE 2031 Final Project — Jan 2026 - May 2026

VHDL, SPI, DE10-Standard FPGA, Quartus Prime, SCOMP Assembly

- Designed a custom VHDL I/O peripheral enabling a 16-bit SCOMP processor to read analog data from an 8 channel, 12 bit LTC2308 ADC over SPI.
- Wrote a 5 state FSM managing SPI timing at 25 MHz, shifting 6 bit channel config words on SDI while capturing 12 bit results on SDO, behind an 8 register memory mapped I/O interface.

Technical Skills

Embedded & Hardware: C (bare-metal), C++, Embedded Firmware, STM32 (ARM Cortex-M), Raspberry Pi Pico (RP2040), TinyUSB / USB HID, VHDL, FPGA (Quartus), SPI, I2C, UART/Serial, PWM, GPIO/Timers, SD Card / FatFS, Arduino, KiCad (PCB Design), Fusion 360, 3D Printing, Oscilloscope, Logic Analyzer

Languages: Python, C, C++, Java, JavaScript, TypeScript, VHDL, Assembly (SCOMP), HTML/CSS

Software & Tools: FastAPI, ReactJS, LiteLLM, OpenCV, Git, Linux, AWS (Lambda, S3, API Gateway), Firebase

Relevant Coursework: Digital Design Lab, Digital Logic Design, Computer Organization & Programming, Systems & Networks, Computer Vision, Machine Learning